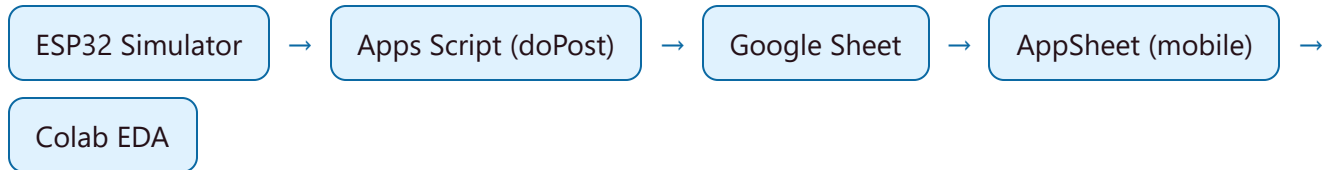


Activity 3 — Build Your Own Smart Mushroom Farm Mobile App

A complete IoT monitoring system: a sensor **simulator** sends data to a **Google Apps Script**, which stores it in **Google Sheets**, displayed as a live mobile app in **AppSheet** — and analyzed in **Colab**.



Prerequisites

- Google account (free), Chrome browser, a smartphone for testing.
- Simulator: <https://setcertification.com/datascience/baguio/session5/> (also [index2.html](#) and [index3.html](#) for the other simulators).

Part 1 — Create your Google Sheet

1. Google Drive → + **New** → **Google Sheets**.
2. Bold + color Row 1 (headers get created automatically by the script).
3. Copy the **Sheet ID** from the URL: `.../spreadsheets/d/[THIS_IS_YOUR_SHEET_ID]/edit` — save it.

Part 2 — Google Apps Script (the backend)

1. Go to script.google.com → **New Project** → name it `MushroomFarmAPI_[YourName]`.
2. Delete the default code and paste the script below.
3. Replace `YOUR_SHEET_ID_HERE` with your Sheet ID, then **Save**.
4. **Deploy** → **New deployment** → **Web app**; Execute as **Me**; Who has access **Anyone**; **Deploy**; authorize.
5. Copy the **Web App URL** (ends with `/exec`) — you'll paste it into the simulator.

This one script **auto-detects** the simulator type (Mushroom Farm, Weather Station, Tourism Monitor, Forest Watch) and sets the right headers automatically.

APPS SCRIPT (CODE.GS)


```

    return
    ContentService.createTextOutput(JSON.stringify(response)).setMimeType(ContentService.MimeType.JSON);
  }
  return ContentService.createTextOutput(JSON.stringify({'message': 'No data available'})).setMimeType(ContentService.MimeType.JSON);
} catch (error) {
  return
  ContentService.createTextOutput(JSON.stringify({'error': error.toString()})).setMimeType(ContentService.MimeType.JSON);
}
}

```

Part 3 — Connect the simulator

1. Open the simulator URL in Chrome (see Prerequisites).
2. In the Control Center, paste your **Web App URL**; set a Device ID; choose Manual / Auto / Stress.
3. Click **Start Simulation** (or **Send Once** to test). Check your Sheet — rows should appear.

No data? Confirm the Web App URL is correct and ends with /exec. Try **Send Once** first.

Part 4 — Build the AppSheet mobile app

1. appsheet.com → sign in → **Create** → **App** → **Start with existing data** → choose your Sheet.
2. Customize: app name **Smart Mushroom Farm Monitor**, theme green, add a logo.
3. Add views: **Dashboard**, **Current Conditions** (detail of latest row), **Temperature Trend** (line chart), **All Readings** (table sorted by Timestamp desc).
4. Format rules: Temperature 18–24 green, outside red. Add a **Status** virtual column (formula below).
5. Automation → Bots: alert when [Temperature] > 28.

APPSHEET — STATUS VIRTUAL COLUMN

```

IF(AND([Temperature]>=18, [Temperature]<=24, [Humidity]>=80, [Humidity]<=95),
  "✅ Optimal", "⚠️ Check Conditions")

```

APPSHEET — OPTIMAL RANGE INDICATOR

```
IF(AND([Temperature]>=18,[Temperature]<=24,[Humidity]>=80,[Humidity]<=95), "●",  
IF(OR([Temperature]<15,[Temperature]>30,[Humidity]<60), "●", "●"))
```

Part 5 — Analyze in Colab

Open the **Activity 3 Colab notebook**. Set SHEET_NAME to your sheet's name and run it — the script auto-detects the simulator type and produces overview stats, time-series and correlation charts, anomaly detection, insights, and an enhanced CSV export. It works for all four simulators.